

Characteristics of Driver Peripheral Vision: How Drivers Respond to Ubiquitous Information on Wide-Area In-Vehicle Displays

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Abstract

Despite advancements in In-Vehicle Information Systems (IVIS) and extensive research on screen layouts, the influence of drivers' peripheral vision on interactions with evolving multi-screen and large display technologies remains poorly understood. This study examines drivers' responses to in-vehicle interactive information through peripheral vision, aiming to optimize visual interaction efficiency and enhance driving safety. Analyzing data from 216 participants in a driving simulator, we explored how horizontal eccentricity, screen type, cognitive load, visual crowding, and stimulus type affect perception rates and reaction times. Our findings highlight the significance of these factors and the need for driver-centered design. The results suggest designing IVIS that align with natural visual tendencies to improve interaction efficiency and driving safety.

CCS Concepts

• Human-centered computing → Empirical studies in HCI; Empirical studies in interaction design.

Keywords

peripheral vision, visual crowding, driver perception, wide-area in-vehicle displays

ACM Reference Format:

Hongwei Huang, Jiateng Li, Xuejing Feng, Jun Ma, and Bruce Mehler. 2025. Characteristics of Driver Peripheral Vision: How Drivers Respond to Ubiquitous Information on Wide-Area In-Vehicle Displays. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719913>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719913>

1 Introduction

Research on the road safety aspects of peripheral vision has spanned several decades. Information acquisition theory states that peripheral vision acts as a medium for drivers' attention shifts and also explains how drivers perceive the world while driving [45]. Information that demands the driver's attention often appears in peripheral vision, the area outside the direct line of sight. This information can be classified into two types: external, such as surrounding traffic and road conditions, and internal, such as vehicle status displays (e.g., speed), visual warning indicators, navigation and entertainment displays. Complex tasks like driving involve serialized decision-making [36], where a driver typically focuses on one task at a time, with attention divided; from a safety perspective, the driver ideally concentrates primarily on the road ahead, but may intermittently engage in internally directed tasks, like interacting with the in-vehicle information systems (IVIS) [7]. As a result, peripheral vision becomes the medium for shifting attention between these tasks. When glancing away from the road, drivers rely on their peripheral vision to control the vehicle, such as maintaining lane position and identifying potential hazards ahead [22, 40]. A similar observation has been made using the visual occlusion technique to assess visual demands while driving [21, 35]; retaining peripheral vision proves essential, facilitating safe lane-keeping even when the direct forward view is obstructed [10, 17, 18, 34].

Extensive research has focused on the placement of IVIS displays [16, 23, 28] and their impact on driving performance [22]. However, as automotive cockpits continue to evolve, with features like larger and multi-screen displays, there is a gap in understanding how these changes affect drivers' peripheral vision. For example, the Lincoln Navigator features a 48-inch wrap-around screen, IM Motors LS6 includes both a 26.3-inch cockpit screen and a 15.5-inch passenger display, and BMW's iDrive system offers panoramic vision across the lower windshield. While these display technologies afford extensive interaction space for peripheral interaction, which is viewed as promising in supporting drivers' situational awareness[9], the potential for information to appear across the entire cockpit raises critical questions: Where should drivers focus? Will their attention be diverted, and can they still effectively perceive critical information? This study aims to help address this gap by investigating

how wide-area in-vehicle displays impact drivers' peripheral vision, conducting an experiment in a driving simulator to identify key factors affecting driver awareness and performance. Employing the Peripheral Detection Task (PDT) and eye tracking, we measured 216 drivers' perception rates and reaction time, while monitoring their visual behavior. This methodology, grounded in vision science, allowed us to analyze the influence of horizontal eccentricity, screen type, cognitive load, visual crowding, and stimulus type.

2 Related Work

Peripheral vision, also known as indirect or side vision, detects objects outside the direct line of sight. This type of vision is crucial for identifying potential dangers and threats, enabling quick reactions to environmental changes. By 1927, the ophthalmologist Traquair utilized a perimeter to map the visual field while studying visual field deficits [41]. Following this, most research on peripheral vision has employed similar methodologies: one typical approach involves having participants respond to artificial stimuli appearing at specific eccentricities [2, 3, 44].

Subsequently, the importance of peripheral vision in driving led to the adoption of the PDT, which periodically presents visual stimuli that briefly appear or change state at one or more predetermined locations that participants are instructed to quickly respond to while maintaining attention on the road. The PDT was first used in a simulator study to evaluate different designs and functionalities of Advanced Driver Assistance Systems (ADAS) [13]. Since then, it has been used primarily as a method to assess driving distractions caused by IVIS or other cognitive load tasks [20, 27, 31]. This work employs the PDT as a method to evaluate the characteristics of drivers' peripheral vision.

Recent trends in automotive cockpits focus on intelligent, large-screen (or multi-screen) designs, with an increasing amount of entertainment and Points of Interest (POI) information displayed. These larger screens provide more space for diverse information within in-vehicle information systems. However, incorrect or complex information may lead to driver perception failures [37], which is particularly dangerous in driving scenarios. A recent study on the copilot screen validates this, showing how drivers can be distracted when copilot passengers watch videos, occasionally drawing the driver's attention to the screen content [25].

Research has demonstrated that cognitive load degrades peripheral vision, causing visual tunneling or decreased sensitivity to peripheral stimuli [32, 33], which impairs a driver's ability to respond to unexpected events. Additionally, when information is densely arranged, visual crowding can interfere with target recognition, causing delays or failures in recognition [6, 39]. While studies have explored the impact of icon size, color, and layout on IVIS performance [5, 11, 30, 47], few have addressed how compact arrangements within the same interface might lead to visual crowding. This paper incorporates this factor into the experimental design to investigate peripheral vision performance under wide-area active IVIS conditions.

3 Method

3.1 Screen Type

Three screen types were configured based on common IVIS display devices (see Fig. 1): Screen 1 (vertical eccentricity: 20° to 25°; focal plane: 70 cm) simulates a traditional combination of the instrument panel and center control screen; Screen 2 (vertical eccentricity: 12° to 20°; focal plane: 90 cm) represents a screen slightly farther from the driver, similar to the surround screen of the Lincoln Nautilus; Screen 3 (vertical eccentricity: 8° to 15°; focal plane: 124 cm) is positioned at the bottom edge of the windshield, akin to some W-HUDs or BMW's Panoramic Vision.

3.2 N-back task

An auditory presentation - verbal response N-back working memory task was used to impose cognitive workload on participants [15, 26]. Given that the car-following task already involves cognitive workload, a moderately demanding 1-back level was used in which single digits (0-9) were presented in random order every 2.5-3.5 seconds and participants were asked to say out loud the digit immediately preceding the current one.

3.3 Stimulus type and visual crowding

Most research on peripheral vision and hazard perception uses a PDT to indirectly measure perception [20], with stimuli typically presented as red dots in simulated driving scenarios to represent unexpected pedestrians or vehicles [31], or as LEDs within the vehicle during real driving conditions [49]. This study focuses on drivers' perception of interactive information, so the stimuli correspond to interactive content, using red dots and red directional arrows to represent alerts and information, respectively (see Fig. 2(a)). The signal stimuli simulate alerts from the IVIS that drivers will ideally notice without requiring direct gaze, such as ambient lights flashing to indicate danger. The information stimuli, designed to test visual crowding, mimic real interactions requiring peripheral vision to assess whether drivers engage visually before responding, such as images from electronic rearview mirrors on the IVIS [24, 29, 48].

We established two conditions commonly associated with visual crowding on IVIS: static and dynamic flankers (see Fig. 2(b)). The static flankers are based on traditional visual crowding experimental setups, utilizing patterns similar to the target stimuli because Bouma's rule applies when the target and flankers are similar [6]. Dynamic flankers reflect the nature of most IVIS content, which is dynamic, with interactive information placed next to constantly changing animations. For signal stimuli (red dots), static flankers were designed to resemble red ABS warning signals, while dynamic flankers represented the driving speed. For information stimuli (red directional icons), static flankers consisted of similar directional arrows, and dynamic flankers again represented the driving speed.

3.4 Peripheral Detection Task (PDT)

For each type of stimulus (signal and information), the PDT had specific performance requirements. For signal stimuli, participants pressed a button on the right side of the steering wheel when they detected the stimulus (see Fig. 3(c)). For information stimuli, participants were to identify the direction of the arrow without direct

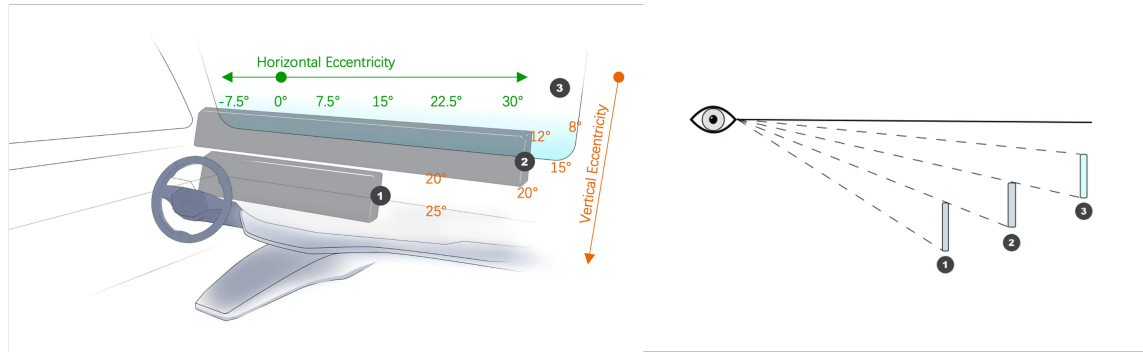


Figure 1: The setup of three types of screen configurations.

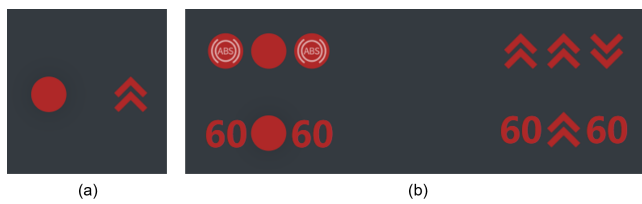


Figure 2: Stimulus type and visual crowding: (a) red dot and red arrow; (b) red dot and red arrow at different visual crowding levels.

gaze and pressed the corresponding directional button. Stimuli were presented at positions of -7.5° , 0° , 15° , 22.5° , and 30° on each of the three screen types, arranged in two vertical rows, totaling 10 points. Prior to the experiment, participants were informed of possible stimulus locations. The experiment began with a cleared background (flankers visible in visual crowding conditions) and randomly appearing stimuli (see Fig.3(a) and (b)). Each stimulus lasted 2 seconds, with random intervals between 13 and 16 seconds.

3.5 Apparatus

Three 43-inch screens displayed the simulated driving environment, which was created using SCANer studio (see Fig.4). The steering wheel and other physical driving controls used Fanatec hardware. Three widescreen monitors were mounted inside the cockpit to represent the three types of IVIS screens. Additionally, the seat was engineered to be adjustable based on participant height, aiming to ensure that the eccentricities of stimuli align with predefined standards. Tobii Glasses 3 were used to monitor in real time whether drivers maintained their visual attention on the car-following task.

3.6 Procedure

Participants were naive to the purpose of the study. They were given 5 minutes to adapt to the simulator and familiarize themselves with the car-following task. The experimenter then explained the 1-back task and the PDT, emphasizing priorities: 1) maintain a proper following distance, 2) ensure accuracy in the 1-back task, and 3) complete the PDT. If the gaze tracking signal from the eye-tracker suggested that a participant was intentionally scanning the PDT locations rather than focusing on the driving or N-back task, they

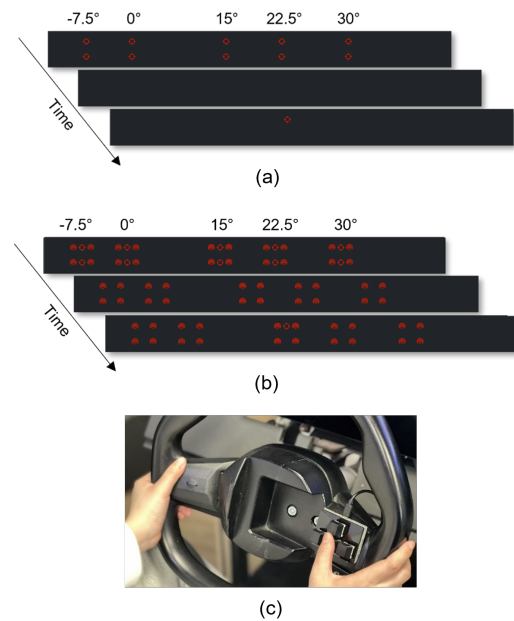


Figure 3: The experimental procedure of PDT. (a) is the procedure under the no visual crowding condition; (b) is the procedure under the visual crowding condition; (c) is the button on the right side of the steering wheel.

were reminded that driving should be their highest priority. The experiment involved two stimulus types (signal and information), three screen types, and three visual crowding conditions, creating 18 combinations. Each session included two conditions: 1) PDT and 2) 1-back task + PDT. To avoid bias, the 216 participants were randomly divided into 18 groups and the order of the tasks was randomized.

In the primary task, a blue box outline is superimposed on the scene to help the driver maintain the proper distance from the vehicle ahead and keep their sight focused straight ahead (0° horizontal and 0° vertical eccentricity, see Fig.5). The size and positioning relative to the lead vehicle adjusts so the box is centered and snugly brackets it at the target following distance, becoming smaller or



Figure 4: The driving simulator.

larger as the following distance is too close or too far, respectively. This setup is based on the traditional useful field of view test [46], where one stimulus is presented centrally and another peripherally. The driver is instructed to maintain a moderate distance, simulating real-world cognitive load while keeping attention on the road. The red halo in the figure represents the driver's gaze point as measured by the Tobii Glasses.

4 Results

4.1 Horizontal eccentricity

The larger the eccentricity, the lower the perception rate and the longer the reaction time, both on the left and right sides (see Fig.6). Large standard deviations suggest that cognitive workload, screen type, visual crowding, and stimulus type may have significant impacts on the outcomes. Therefore, the following analysis will separately examine their effects on perception rate and reaction time.

4.2 Cognitive workload

The results of the Kruskal-Wallis test show that cognitive workload has a significant effect on perception rate ($H = 280.76, p < 0.001$), with a noticeable reduction at high levels of cognitive workload. For example, perception rate dropped by 18.17% (from 60.06% to 41.89%) at 30°, and by 13.68% (from 87.72% to 74.04%) at 0° (see Fig.7). Cognitive workload also significantly affected reaction time, with a significant difference observed ($H = 29.34, p < 0.001$). For instance, reaction time increased by 99 ms (from 1217 ms to 1316 ms) at 30°, and by 62 ms (from 1034 ms to 1096 ms) at 0°. As eccentricity increases, the impact of cognitive load on both perception rate and reaction time becomes more pronounced. It is noteworthy that at a horizontal eccentricity of -7.5°, the average perception rate is lower, and the average reaction time is higher than at 15°. This phenomenon will be discussed in detail in the Discussion section.

4.3 Screen Type

We evaluated the impact of different screen types on participants' perception rate and reaction time. Significant differences were detected using the Kruskal-Wallis test for both perception rate ($H = 311.44, p < 0.001$) and reaction time ($H = 321.76, p < 0.001$), justifying further pairwise comparisons. Dunn's post-hoc tests



Figure 5: The vehicle in front at different distances: from top to bottom, they represent moderate, far, and near.

with Bonferroni correction were conducted to determine specific differences between screen types. These results confirm that different screen types have significantly different impacts on both how quickly and accurately participants responded to stimuli. Unlike Screen 1 and 2, which show a similar trend of initially increasing and then decreasing perception rates with increasing horizontal eccentricity, Screen 3 exhibits a distinct trend with a notable dip at 15 degrees. The underlying reasons for this deviation will be elaborated in the Discussion section (see Fig.8).

4.4 Visual crowding

The results show significant differences in perception rate ($H(2) = 487.87, p < 0.001$) and reaction time ($H(2) = 265.38, p < 0.001$) across different levels of visual crowding. Duncan's multiple range tests elucidated specific pairwise differences. For perception rate, significant reductions were observed when comparing no flankers (Level 0) to static flankers (Level 1; mean difference = -0.2458, $p < 0.001$) and dynamic flankers (Level 2; mean difference = -0.2155, $p < 0.001$). Interestingly, dynamic flankers (Level 2) exhibited a slightly higher perception rate compared to static flankers (Level 1;

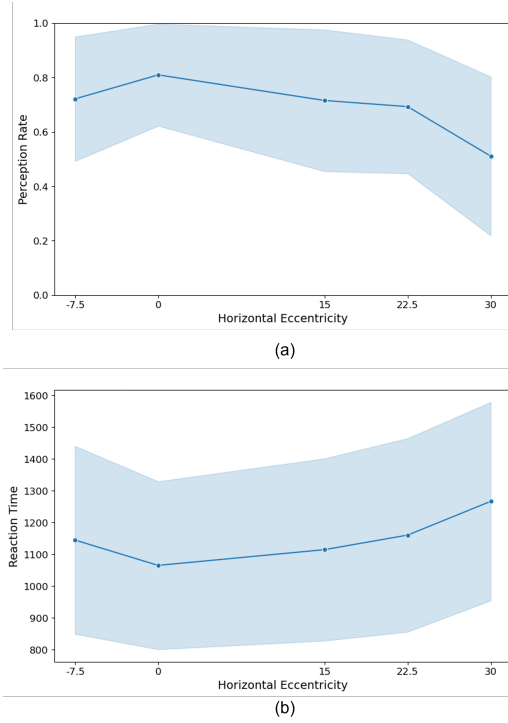


Figure 6: Perception rate (a) and reaction time (b) corresponding to eccentricity.

mean difference = 0.0303, $p < 0.001$). For reaction time, increasing visual crowding from no flankers (Level 0) to static flankers (Level 1) led to a marked increase (mean difference = 232.193 ms, $p < 0.001$). Although reaction times were also longer with dynamic flankers (Level 2) compared to no flankers, they were slightly reduced when compared to static flankers (mean difference = -57.629 ms, $p < 0.001$). These results demonstrate that visual crowding significantly affects both perception rate and reaction time. The presence of static flankers causes more pronounced delay and impairment than dynamic flankers (see Fig. 9).

4.5 Stimulus type

The analysis revealed significant differences in perception rate across the various stimulus types ($H = 79.33$, $p < 0.001$), indicating that the type of visual stimulus significantly influences how accurately subjects are able to perceive stimuli. For example, the perception rate dropped by 17.57% (from 59.83% to 42.26%) at 30°, and by 5.9% (from 83.89% to 77.99%) at 0° (see Fig. 10). Further analysis of reaction times also demonstrated significant variability dependent on the stimulus type ($H = 963.02$, $p < 0.001$). For example, reaction time increased by 397 ms (from 1068 ms to 1465 ms) at 30°, and by 326 ms (from 902 ms to 1228 ms) at 0°. As the eccentricity increases, the impact of cognitive load on both perception rate and reaction time becomes more pronounced. It is noteworthy that at a horizontal eccentricity of -7.5° the average perception rate is lower and the average reaction time is higher than at 15°. This phenomenon will be discussed in detail in the Discussion section. This substantial

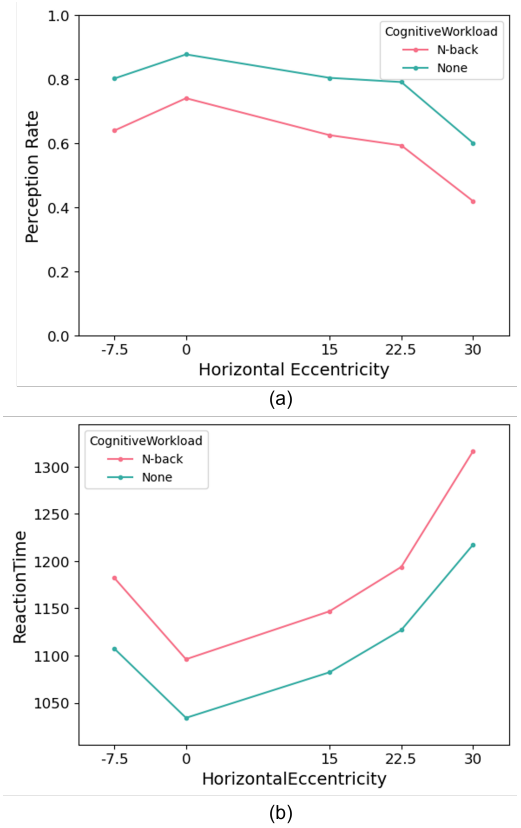


Figure 7: Perception rate (a) and reaction time (b) across the visual field at different cognitive workload levels.

difference suggests that the stimulus type profoundly affects the speed at which participants respond, highlighting the critical role of stimulus type in cognitive processing speed.

5 Discussion

The results show that the highest perception rate and shortest reaction time occur at 0° eccentricity. As eccentricity increases, perception rates decrease and reaction times increase, indicating reduced interaction efficiency with the IVIS. A significant drop in perception and a sharp increase in reaction time are observed beyond 22.5°. These findings align with previous research on peripheral vision. Additionally, cognitive workload, screen type, visual crowding, and stimulus type were found to significantly affect interaction efficiency, further advancing research on the impact of spatial design, interface layout, and information types on peripheral vision within wide-area IVIS.

Regarding horizontal eccentricity, the results revealed a lower perception rate and a longer reaction time at -7.5° compared to 15°. This finding does not entirely align with previous studies that analyzed left and right peripheral vision as symmetrical and concluded that the perception rate decreases and reaction time increases with greater eccentricity [49]. Studies have shown that the neural processing mechanisms in the retina and brain may lead to asymmetry

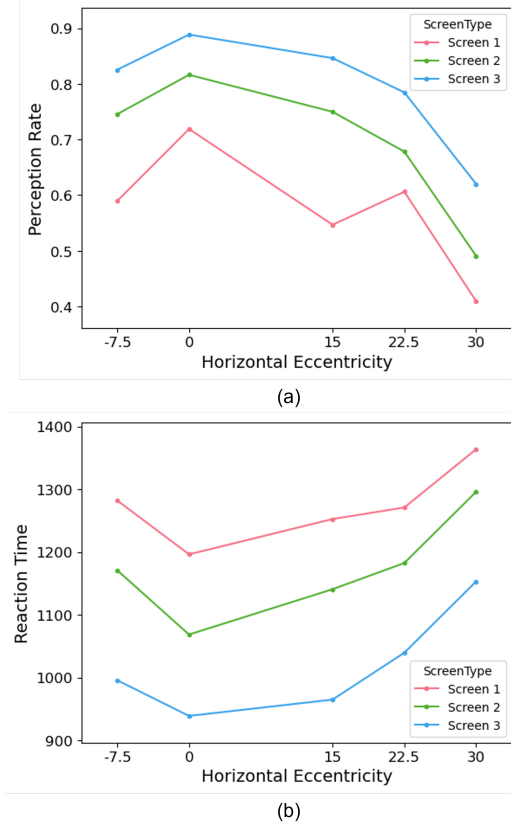


Figure 8: Perception rate (a) and reaction time (b) across the visual field at different screen types.

in the visual field at the physiological level [4, 14]. However, the horizontal asymmetry in the visual field observed here may be attributed to behavioral factors. In left-hand driving scenarios, the layout of IVIS is typically centered around the vehicle's central axis. For the driver, this means that the system's central area is on the right side, with more interactive information displayed. As a result, the driver's attention resource is pre-allocated to the right, leading to a lower perception rate of information on the left. This phenomenon may also be considered a form of spatial attention bias [1, 8, 12]. When designing IVIS, we can leverage this spatial attention bias by displaying frequently used and urgent information on the driver's right side, enabling quicker response times to such information.

It is noteworthy that at 15° on screen 3, the perception rate dropped significantly (54.67% at 15° vs. 60.61% at 22.5°), which contrasts sharply with the trend observed on screen 2 and screen 1 (see Fig. 5). The eye-tracker videos revealed that the 15° eccentricity on screen 3 is located near the steering wheel. A possible explanation for this observation could be the visual adaptation effect, which refers to the gradual decrease in the visual system's response to sustained or repetitive stimuli [19, 42]. When an object remains in a fixed position for an extended period, especially in a near-focal area, the human eye may gradually "adapt" to its position and appearance. This adaptation makes the object and the surrounding area

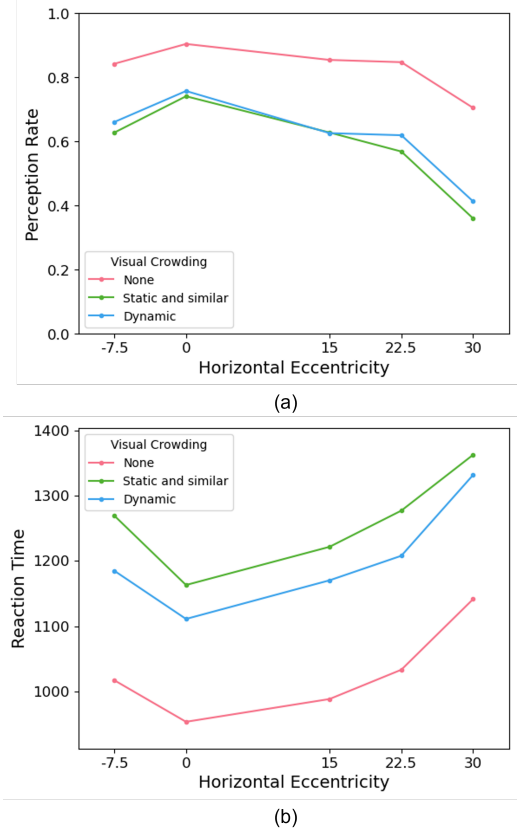


Figure 9: Perception rate (a) and reaction time (b) across the visual field at different visual crowding levels.

less conspicuous, leading to less efficient processing by the visual system. Consequently, although the eccentricity here is relatively small, the perception rate is diminished due to visual adaptation. Therefore, critical information should be avoided in these areas.

While this study highlights the effects of large eccentricities on perception rates and reaction times, it does not suggest that IVIS design should neglect these areas or reduce display ranges. Placing some types of display content at large eccentricities may make sense in meeting some market demands and should be considered, while small eccentricities can also be affected by visual crowding or high cognitive load and critical content placement situated accordingly. A Previous study on copilot screens has shown that displaying videos can distract the driver [25]. However, by utilizing the characteristics of peripheral vision combined with calm technology [43], visual adaptation to copilot screen content can reduce distraction, as it only shifts information to central attention when necessary. As noted, visual crowding can occur even at small eccentricities or central vision, not just in the periphery [38]. Therefore, recognizing and fully leveraging the characteristics of peripheral vision is crucial for the user-friendly design of IVIS and enhancing safety.

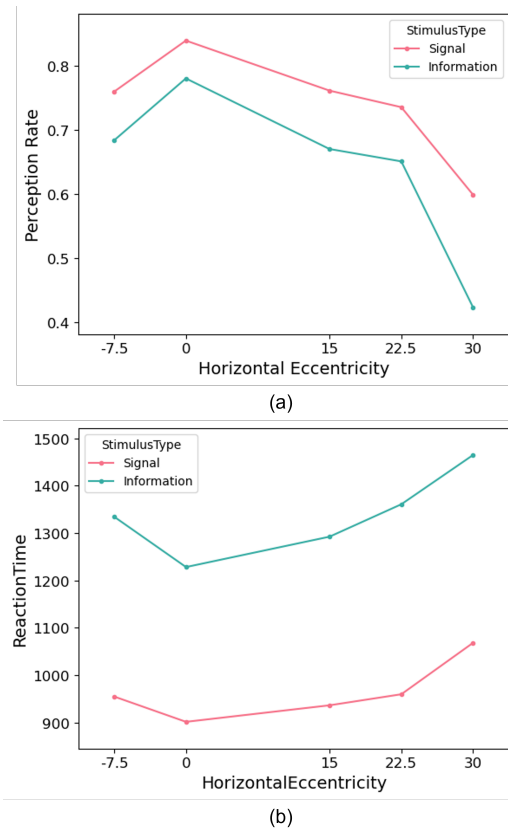


Figure 10: Perception rate (a) and reaction time (b) across the visual field at different stimulus types.

6 Conclusion

This study's findings highlight the characteristics of drivers' peripheral vision and their perceptual performance with active IVIS in driving scenarios. It demonstrates an asymmetry in drivers' peripheral vision, with perceptual performance generally decreasing as visual eccentricity increases. Additionally, factors such as high cognitive load, visual crowding, and stimulus type significantly impact perceptual performance. These insights can play an important role in enhancing the design of wide-area IVIS to improve both usability and driving safety.

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